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Polarized neutron diffraction with *ex-situ* ^{3}He neutron spin filter and two-dimensional detector for a complex incommensurate magnetic order

S. Takahashi,[1,2] Y. Osawa,[1] R. Kobayashi,[2] T. Ino,[3] H. Saito,[1] T. Kawasaki,[2] T. Oku,[2,4] T. Nakamura,[2] Y. Ikeda,[5] M. Fujita,[5] Y. Tokunaga,[6] T. Arima,[6,7] and T. Nakajima[1,3,7]

[1] *Institute for Solid State Physics, The University of Tokyo, Kashiwa, Chiba 277-8581, Japan*
[2] *J-PARC Center, Japan Atomic Energy Agency, 2-4 Shirakata, Tokai, Ibaraki 319-1195, Japan*
[3] *Institute of Materials Structure Science, High Energy Accelerator Research Organization, 1-1 Oho, Tsukuba, Ibaraki 305-0801, Japan*
[4] *Ibaraki University, 2-1-1 Bunkyo, mito, Ibaraki 310-8512, Japan*
[5] *Institute for Materials Research, Tohoku University, 2-1-1 Katahira, Sendai 980-8577, Japan*
[6] *Department of Advanced Materials Science, University of Tokyo, Kashiwa 277-8561, Japan*
[7] *RIKEN Center for Emergent Matter Science, Wako, Saitama 351-0198, Japan*

Polarized neutron scattering has been widely used for studying magnetic orders in condensed matter. One of the typical setups for this technique is a triple-axis spectrometer with ferromagnetic single-crystal monochromator and analyzer, which is suited for point-by-point measurements on a horizontal scattering plane. However, this setup cannot fully meet the demands of recent studies on complex magnetic orders with emergent cross-correlated phenomena. For instance, multiferroics, magnetic skyrmions and other topologically-nontrivial magnetic orders tend to have incommensurate magnetic modulation vectors running along various directions in the reciprocal space, which could not be captured by measurements restricted to the horizontal scattering plane. To overcome these limitations, we developed an experimental method combining a ^{3}He neutron spin filter (^{3}He-NSF) with a two-dimensional (2D) detector, which enables polarization analysis of out-of-plane scattering over a wide Q region. This method was implemented on the polarized neutron triple-axis spectrometer PONTA at JRR-3 and applied to single-crystal diffraction measurements on the multiferroic material Mn_3WO_6, which exhibits complex incommensurate magnetic orders. Polarization-dependent intensities of the incommensurate magnetic reflections were successfully observed and converted into three-dimensional distributions of spin-flip and non-spin-flip intensities in reciprocal space with the correction for the time-varying polarization of ^{3}He nuclear spins. These results demonstrate the effectiveness of the present method for analyzing complex magnetic structures.

I. INTRODUCTION

Polarized neutron scattering is a powerful technique for investigating magnetic structures and spin correlations because of the intrinsic spin and magnetic moment of neutrons [1]. In particular, longitudinal polarization analysis, which analyzes the neutron spin states before and after scattering enables the separation of magnetic and nuclear scattering, the separation of coherent and incoherent scattering, and the determination of magnetic moment directions [2]. In reactor-based neutron experiments, such polarization analysis is typically performed using the (111) plane of the ferromagnetic crystal Cu_2MnAl as the neutron polarizer and analyzer at triple-axis spectrometers[3–7], as shown in Fig. 1(a). While this type of instrument has been widely used for magnetic structure analysis, the accessible Q region for analyzing scattered neutrons is limited to the scattering plane. Therefore it has been mainly applied to relatively simple magnetic structures.

In recent years, topologically nontrivial magnetic textures, such as magnetic skyrmions and spin-hedgehog lattices, have attracted considerable attention because their complex spin arrangements can give rise to emergent electromagnetic responses and other unconventional physical phenomena [8, 9]. These magnetic orders are often characterized by a superposition of multiple magnetic modulation wavevectors (multiple q-vectors). As a result, the corresponding magnetic reflections are not confined to a single scattering plane but are distributed over three-dimensional reciprocal space.

Since the discovery of the magnetic skyrmion lattice in MnSi by small-angle neutron scattering (SANS) [8], SANS has played a central role in identifying long-period magnetic structures by detecting their magnetic modulation wavevectors [10–12]. In addition, polarized SANS, which is combined with a large-solid-angle neutron spin analyzer, such as using polarized ^{3}He (^{3}He neutron spin filter, ^{3}He-NSF) has become a powerful technique for analyzing such complex magnetic structures [13, 14]. However, the accessible Q range in SANS experiments is generally limited to relatively low Q. To investigate broader range of complex magnetic structures, it is desirable to extend polarization analysis to higher-Q regions. Thus, we propose a polarization analysis technique in which a ^{3}He-NSF and a two-dimensional detector are placed at arbitrary scattering angles, as shown in Fig. 1(b). Note that this type of instrument has already been reported at several neutron facilities, such as the POLI at MLZ [15], and WOMBAT at ANSTO [16]. However, to the best of our knowledge, quantitative experimental results demonstrating polarization analysis with this type of setup have not yet been reported.

In this work, we implemented polarization analysis using a ^{3}He-NSF combined with a two-dimensional position-sensitive detector on the Polarized neutron triple axis spectrometer, PONTA at JRR-3. Polarized neutron diffraction experiments on the multiferroic material Mn_3WO_6 was performed using this setup. The performance of the ^{3}He-NSF was evaluated from the obtained nuclear reflection data. Furthermore, three-dimensional reciprocal-space mapping data are presented to show that the present method enables polarization analysis of complex magnetic structures distributed over three-dimensional reciprocal space.

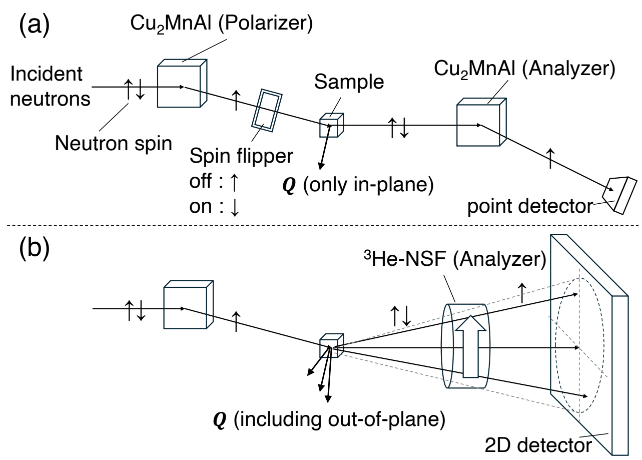

FIG. 1. Schematic illustrations comparing neutron polarization analysis techniques. (a) Conventional setup using a Cu_2MnAl crystal for both polarizer and analyzer, where only in-plane scattered neutrons are detected by a point detector. (b) Polarization analysis setup using a ^{3}He-NSF combined with a two-dimensional detector, enabling detection of both in-plane and out-of-plane scattered neutrons.

II. EXPERIMENTAL DETAILS

A. Setup

PONTA is one of the polarized neutron diffractometer in JRR-3, and it has a longitudinal polarization analysis option [7]. In this option, both incident and scattering neutrons are polarized and analyzed by Cu_2MnAl Heusler crystal monochromator, which provides a neutron wavelength of 2.3 Å. In this experiment, the spin analyzer of the Cu_2MnAl Heusler crystal monochromator and point detector were not used, and ^{3}He-NSF and two-dimensional detector were installed instead. Figure 2 shows a overview of the experimental setup from the sample to the detector. The neutron detector was a scintillation-type two-dimensional detector with an active area of 256 mm × 256 mm using wavelength-shifting fiber [17] . The detector and ^{3}He-NSF were positioned at an angle of approximately 32° relative to the incident neutron beam direction and fixed during the measurement. The sample-to-detector distance was approximately 0.9 m. To maximize the solid-angle coverage, the ^{3}He-NSF was placed as close to the sample as possible. The vertical angular coverage of the ^{3}He-NSF was 4.7°, enabling polarization analysis over approximately half of the detector area around the detector center. To suppress the background level, a neutron guide tube covered with B_4C shielding tube was installed between the ^{3}He-NSF and the detector. A single-crystal sample of Mn_3WO_6 with a volume of 8 mm^3 was mounted on an aluminum holder and installed in a cryostat with the scattering plane set to the $(H0L)$ plane. The rotation angles ω of the sample were adjustable to align the target reflection with the detector. The Helmholtz coil was used to maintain the neutron spin polarization by applying a vertical magnetic field of approximately 1 mT. The magnetic field in the ^{3}He-NSF was oriented parallel to the beam axis, and spins of the neutrons scattered from the sample adiabatically followed the field direction as illustrated in Fig. 2.

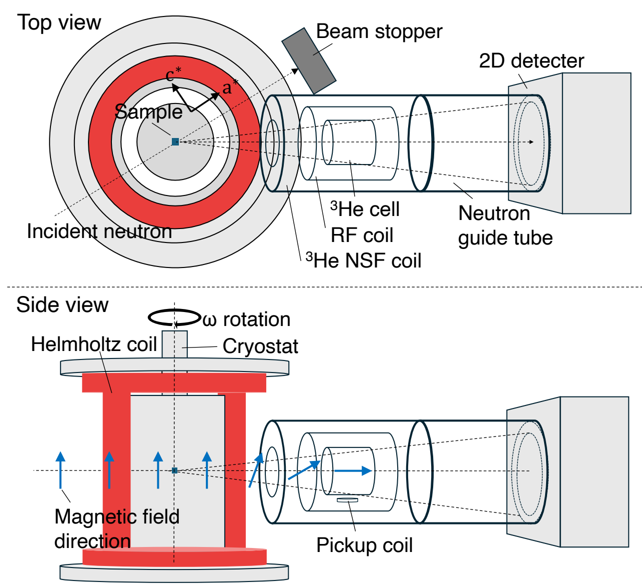

FIG. 2. Schematic illustration of a polarization analysis method enabling broad coverage of reciprocal lattice points in three-dimensional reciprocal space.

B. ^{3}He neutron spin filter

The neutron polarization principle of the ^{3}He-NSF is based on the strong spin dependence of the neutron absorption cross section of ^{3}He nuclei. Neutrons with spins antiparallel to the ^{3}He nuclear spins are strongly absorbed, while those with spins parallel to the ^{3}He nu-

TABLE I. Specifications of ^{3}He cells.

Name	$D \times L$ (mm)	Volume (cm^3)	^{3}He density (amg)	N_2 pressure (Torr)	K/Rb (mol)
Kenji	72×67	140	3.2	97	26
Shimpachi	72×65	146	3.1	100	37

clear spins are transmitted. Thus, neutron polarization requires ^{3}He gas polarization. The ^{3}He polarization was achieved by using Spin-Exchange Optical Pumping (SEOP) method [18]. In this method, circularly polarized laser light polarizes alkali-metal atoms, and the polarization is then transferred to ^{3}He nuclei through hyperfine interactions. The SEOP system employed in this study was designed based on that developed for the polarized neutron spectrometer POLANO at J-PARC MLF [19]. The SEOP system and the polarization analysis coil are equipped with a adiabatic fast-passage nuclear magnetic resonance (AFP NMR) system to flip the ^{3}He polarization [20]. An NMR pickup coil is also placed at the bottom of the ^{3}He cell to monitor the ^{3}He polarization with the NMR signal (Fig. 2). Two ^{3}He glass cells, named Kenji and Shimpachi, made of aluminosilicate glass (GE180) with nearly identical dimensions were used in the experiment. Both cells are filled with ^{3}He gas, nitrogen gas, and a mixture of alkali-metal atoms Rb and K for the hybrid SEOP [21, 22]. The details of the cell specifications are described in Table I. An darkroom for using the SEOP laser was constructed in the beamline experimental area. The ^{3}He gas was polarized in the darkroom. After the saturation, the ^{3}He cell was transported to the beamline, where neutron measurements were performed during the relaxation of the ^{3}He polarization. It is known as an *ex-situ* ^{3}He-NSF method. Because the ^{3}He polarization decreases with time in this method, one cell was used for the polarized neutron experiment while the other was polarized in the SEOP system, and the two cells were alternately exchanged every two or three days during the experiment.

III. RESULTS

In this section, we present the results of polarized neutron diffraction experiments on Mn_3WO_6 performed to demonstrate polarization analysis using a ^{3}He-NSF combined with a two-dimensional detector. Figure 3 shows the procedures for data acquisition and data reduction of polarization-analysis results for magnetic reflections distributed in three-dimensional reciprocal space. The data were acquired by scanning the sample rotation angle ω. For each measurement, the ^{3}He polarization was flipped by AFP NMR to switch the analyzer condition between I^{++} and I^{+-}. Because an *ex-situ* ^{3}He-NSF was employed, ^{3}He polarization was gradually decayed during the experiment, the neutron polarization and transmission of the ^{3}He-NSF also decreased with time. Therefore, corrections for these time-dependence are nesessary.

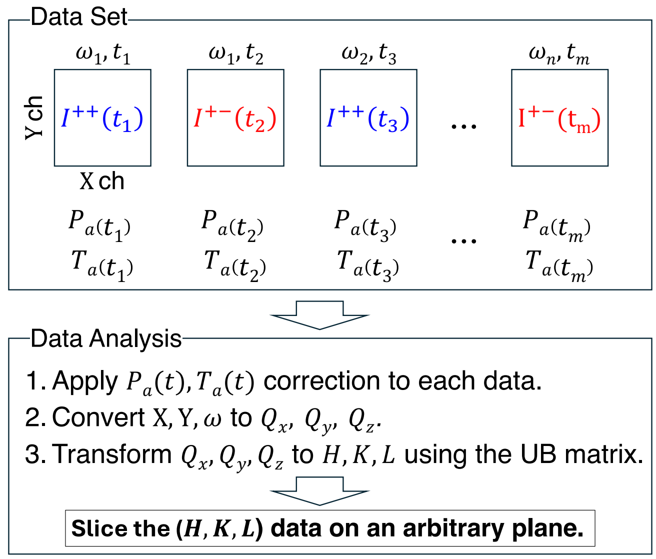

FIG. 3. Schematic diagram of the data-reduction procedure for polarization analysis using an *ex-situ* ^{3}He-NSFand a two-dimensional detector. Raw two-dimensional detector data measured at different sample rotation angles ω_i and times t_i are corrected using the time-dependent neutron polarization $P_n(t_i)$ and transmission $T_n(t_i)$. The detector coordinates (X,Y) and sample rotation angle ω are converted to Q_x, Q_y, Q_z, and then transformed to H, K, L using the UB matrix. The corrected three-dimensional reciprocal-space data can then be sliced on an arbitrary plane.

For this correction, the time dependence of the ^{3}He polarization was evaluated from the nuclear Bragg reflection. After applying these time-dependence correction, the detector coordinates and sample rotation angle were transformed into reciprocal-space coordinates.

A. Evaluation of ^{3}He neutron spin filter from nuclear scattering

The ^{3}He polarization was estimated from the nuclear Bragg peak intensity of Mn_3WO_6 single crystal sample. Nuclear peak intensities where the incident neutrons polarized by Cu_2MnAl single crystal monochromator and scattering neutrons analyzed by ^{3}He-NSF were given by the following equations:

$$I^{+\pm}(t) = N_0 \Sigma^{++} T_p T_a(t)(1 \pm P_p P_a(t)) \qquad (1)$$

where the supersucripts of $I^{+\pm}$ indicate the incident neutron polarization state and the scattered neutron polar-

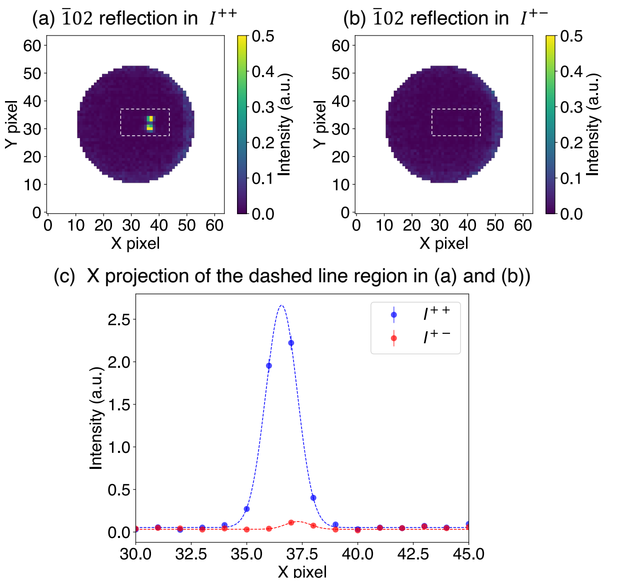

FIG. 4. (a) Non-spin-flip intensity (I^{++}) and (b) spin-flip intensity (I^{+-}) maps, showing only the region covered by the ^{3}He cell. A clear difference in the intensity of the $\bar{1}02$ nuclear Bragg peak is observed depending on the direction of the ^{3}He polarization. (c) Line profile along the (x) direction obtained by integrating the intensity within the dashed regions in (a) and (b).

ization state, respectively, N_0 is a number of incident neutrons comes from the reactor source, Σ^{++} represents the scattering cross section of the $\bar{1}02$ nuclear reflection of Mn_3WO_6, T_p is the neutron transmission of the Cu_2MnAl polarizer, $T_a(t)$ is the neutron transmission of the ^{3}He-NSF, P_p is the neutron polarization of the Cu_2MnAl polarizer, and $P_a(t)$ is the neutron polarization of the ^{3}He-NSF. By summing I^{++} and I^{+-} in Eq 1, the following expression is obtained:

$$I^{++}(t) + I^{+-}(t) = 2N_0\Sigma^{++}T_pT_a(t), \tag{2}$$

Here, the measurement time is sufficiently short compared with the relaxation time of the polarized ^{3}He gas, $I^{++}(t)$ and $I^{+-}(t)$ can be regarded as being measured at the same time. The neutron transmission $T_a(t)$ of ^{3}He-NSF, the neutron polarization $P_a(t)$ of ^{3}He-NSF, and ^{3}He polarization $P_{\mathrm{He}}(t)$ are given by the following equations [23],

$$T_a(t) = T_{\mathrm{cell}}\exp(-\rho d\sigma(\lambda))\cosh(\rho d\sigma(\lambda)P_{\mathrm{He}}(t)), \tag{3}$$
$$P_a(t) = \tanh(\rho d\sigma(\lambda)P_{\mathrm{He}}(t)), \tag{4}$$
$$P_{\mathrm{He}}(t) = P_0\exp(-t/\tau), \tag{5}$$

where T_{cell} is the transmission of the ^{3}He glass cell, ρ is a number density of ^{3}He , d is a gas thickness of ^{3}He , $\sigma(\lambda)$ is a neutron absorption cross section of unpolarized ^{3}He , P_0 is the initial polarization at $t = 0$, and τ is the relaxation time. The wavelength λ of the incident neutrons was monochromatized to 2.36 Å by Cu_2MnAl. By taking the ratio with the intensity I_{unpol} measured when the ^{3}He polarization is zero ($P_{\mathrm{He}} = 0$), the following equation is obtained,

$$\frac{I^{++}(t) + I^{+-}(t)}{I_{\mathrm{unpol}}} = 2\cosh(\rho d\sigma(\lambda)P_{\mathrm{He}}(t)). \tag{6}$$

The ^{3}He gas thickness ρd was determined from the transmission ratio between the ^{3}He cell and an empty cell of the same size. The value of ρd was 18.5 ± 0.5 [amg cm] for Kenji cell and 18.8 ± 0.4 [amg cm] for Shimpachi cell. Note that the amagat is a non-SI unit of the volumetric number density of an ideal gas at 0 °C (273.15 K) and 1 atm (101.325 kPa), and one amagat is equivalent to $2.69 \times 10^{19}/\mathrm{cm}^3$.

Figure 4 (a,b) shows two-dimensional intensity maps measured just after transporting the Shimpachi cell ($P_{\mathrm{He}}(0)$), where the $\bar{1}02$ nuclear reflection intensity clearly changes depending on the I^{++} and I^{+-}. Figure 4 (c) shows the result of projecting the peak region of these intensity maps along the x-axis. The peak intensities for I^{++} and I^{+-} were integrated, and I_{unpol} was also integrated in the same region before polarizing ^{3}He by SEOP. The ^{3}He polarization was then determined using Eq. 6, yielding $P_{\mathrm{He}} = 0.81 \pm 0.05$ for the Shimpachi cell. Figure 5 shows the time evolution of the ^{3}He polarization determined from the nuclear reflection (open circles) and the NMR signal (closed circles). Throughout the experiment, nuclear scattering measurements were periodically conducted between magnetic scattering measurements. The time evolution of the nuclear reflection intensities and the NMR signal showed good agreement and the relaxation time τ was estimated to be $\tau = 131.1 \pm 0.7$ hour. Figure 5 also shows the neutron polarization and transmission calculated from eq. 5 using Shimpachi cell. Also, the same measurement and analysis were performed for the Kenji cell,yielding $P_{\mathrm{He}}(0) = 0.72 \pm 0.06$ and $\tau = 128.0 \pm 0.3$ hour. P_p is also required for the polarization correction of the magnetic scattering data. The asymmetry between I^{++} and I^{+-} is described by Eq. 1,

$$\frac{I^{++} - I^{+-}}{I^{++} + I^{+-}} = P_pP_a(t). \tag{7}$$

Since $P_a(t)$ is independently determined from the ^{3}He polarization, P_p can be obtained and the average value of P_p was found to be 0.86 ± 0.06. Note that P_p was determined for each nuclear reflection measurement and includes the effects of environmental background variations and neutron depolarization in the guide field in the analysis. The validity of the polarization correction was confirmed using the ω scan of $\bar{1}02$ reflection. The integrated intensity at each ω was obtained by integrating the intensity within the peak region identified on the two-dimensional detector in Fig. 4. The raw intensity profiles of the $\bar{1}02$ reflection show a finite spin-flip intensity owing to the imperfect neutron polarization, as shown in Fig. 6(a). By applying the polarization correction, the neutron polarization and transmission were corrected and the non-spin flip and spin-flip scattering cross-sections, Σ^{++} and Σ^{+-}, are obtained as shown in Fig. 6(b). Since only the non-spin-flip scattering cross section Σ^{++} is expected for nuclear scattering, the result indicates that the polarization correction was successfully performed. Details of the polarization correction are described in the Appendix A. These values were used for the polarization correction of the magnetic scattering data.

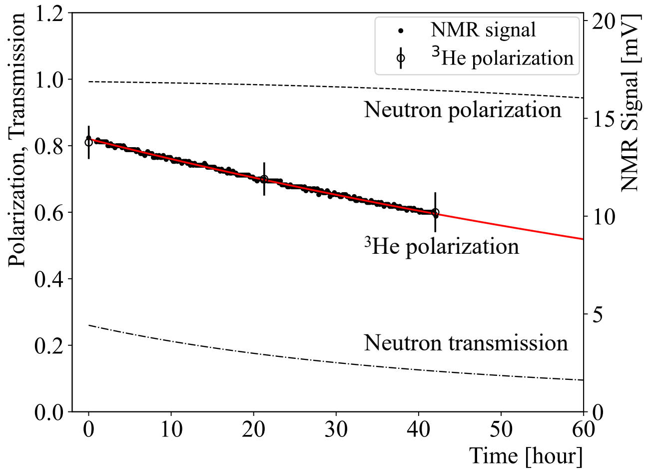

FIG. 5. Time dependence of the ^{3}He polarization, neutron polarization, and neutron transmission. The open circles show the ^{3}He polarization determined from the intensity of the $\bar{1}02$ nuclear reflection, while the closed circles show the ^{3}He NMR signal. The red line represents the fitting result using an exponential decay function. The dashed and dot-dashed lines indicate the calculated neutron polarization and transmission, respectively, obtained from the ^{3}He polarization.

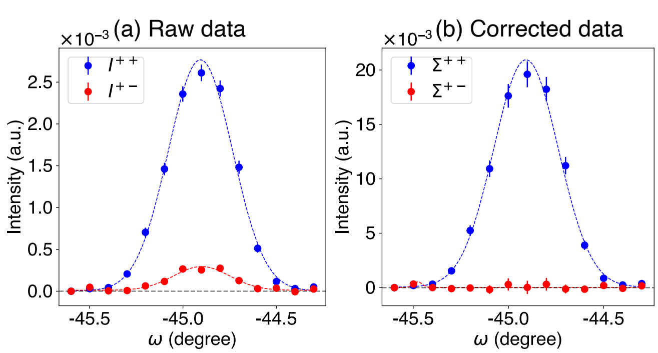

FIG. 6. The measured intensity of the $\bar{1}02$ reflection integrated over the peak region identified on the two-dimensional detector while scanning the sample rotation angle ω. (a) Raw data, where the red and blue symbols represent the non-spin-flip intensity I^{++} and spin-flip intensity I^{+-}, respectively. (b) Corrected data, where the red and blue symbols represent the corrected non-spin-flip and spin-flip scattering cross sections, Σ^{++} and Σ^{+-}, respectively.

B. Polarization analysis of magnetic scattering

Mn_3WO_6 is a multiferroic material with a ferroaxial crystal structure belonging to the R$\bar{3}$ space group [24]. This system exhibits successive magnetic phase transitions at T = 25 K and 22.5 K, and spontaneous electric polarization has been reported to appear in the low-temperature magnetically ordered phase. In this paper, we focus on the polarization analysis results obtained at 23 K, just below the antiferromagnetic transition. First, using unpolarized neutrons and a two-dimensional detector, intensity maps of antiferromagnetic reflections distributed in the three-dimensional reciprocal space were measured. This measurement was performed using the setup shown in Fig. 2 without the ^{3}He-NSF, and the Cu_2MnAl polarizer was replaced by a PG monochromator. Figure 7(a) shows the reciprocal-space region measured in the present experiment. Figure 7(b) and (d) show intensity maps obtained by integrating the measured intensity along the L direction and the $(-K/2, K, 0)$ direction, respectively. Because the scattering plane was set to the $(H0L)$ plane, a conventional triple-axis setup would mainly provide a reciprocal-space map in this plane, as shown in Fig. 7(d). In the present experiment, the use of a two-dimensional detector enabled the measured three-dimensional data to be projected along different directions. In this projection, magnetic reflections characterized by the three propagation vectors $\boldsymbol{q}_1$, $\boldsymbol{q}_2$, and $\boldsymbol{q}_3$ extending from the $\bar{1}02$ reflection are clearly observed. Polarized neutron scattering measurements were then performed over the same reciprocal-space regions. Figures 7(c) and 7(e) show the polarization-analysis maps obtained from the intensity difference $I^{++} - I^{+-}$. Clear positive and negative contrasts are observed at the magnetic reflection positions identified in the unpolarized intensity maps. In addition, to compare the spin-dependent intensities more quantitatively, line profiles along the $-H$ direction through $\boldsymbol{q}_3$ and $\boldsymbol{q}_1$ are shown in Figs. 7(f) and 7(g), respectively. The relative intensities of the non-spin-flip and spin-flip components are different for the two reflections clearly. Note that the data in Figs. 7(c)-(g) are shown after applying the polarization correction described above. This indicates that the spin-dependent scattering intensities were successfully resolved for magnetic reflections distributed in three-dimensional reciprocal space. The polarization-analysis data provide constraints on the magnetic moment components associated with each propagation vector through the relative non-spin-flip and spin-flip intensities. A detailed magnetic structure analysis based on these constraints will be presented in a separate paper.

IV. SUMMARY AND OUTLOOK

In this work, we developed a polarization analysis method that combines a ^{3}He-NSF with a two-dimensional detector to analyze arbitrary reciprocal-

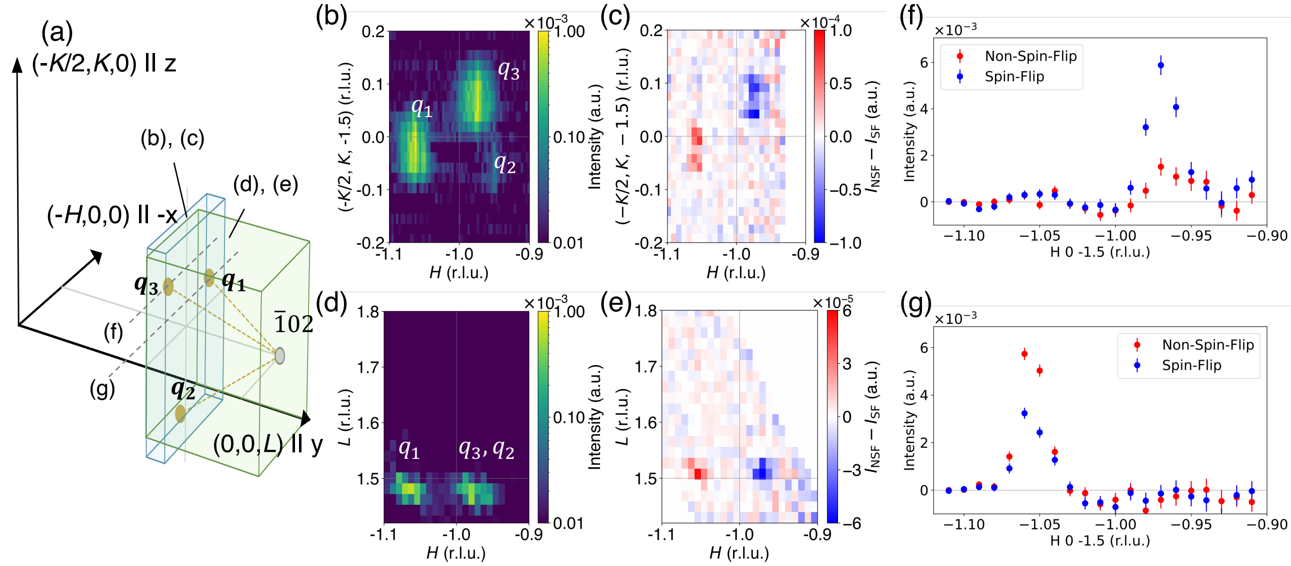

FIG. 7. (a) Schematic illustration of the reciprocal-space region measured in the present experiment. The translucent volumes labeled (b), (c) and (d), (e) indicate the regions integrated along the L direction and the $(-K/2, K, 0)$ direction, respectively. The dashed lines along the $-H$ direction through $\boldsymbol{q}_3$ and $\boldsymbol{q}_1$ indicate the line cuts shown in panels (f) and (g), respectively. (b), (d) Unpolarized neutron intensity maps in the $(H, K/2, -1.5)$ and $(H, 0, L)$ planes, respectively. (c), (e) Corresponding polarization-analysis maps obtained from the difference between the non-spin-flip and spin-flip intensities, $I^{++} - I^{+-}$. (f), (g) Line profiles of the non-spin-flip and spin-flip intensities at $\boldsymbol{q}_3$ and $\boldsymbol{q}_1$, respectively.

lattice points distributed in three-dimensional reciprocal space. Polarization analysis was successfully performed for the incommensurate magnetic reflections of Mn_3WO_6 at 23 K, and the detail of the magnetic structure will be reported in a separate paper. The positions of the ^{3}He-NSF and the two-dimensional detector were fixed in the present experiment, whereas a configuration with both devices mounted on a scanning arm is planned to cover a wider region of reciprocal space. In addition, a compact *in-situ* ^{3}He-NSF system has recently been developed at J-PARC, in which SEOP is continuously performed during neutron scattering experiments to maintain stable neutron polarization [25]. This implementation is expected to provide stable neutron polarization during experiments, extend polarization analysis over a wider region of three-dimensional reciprocal space, and contribute to the elucidation of complex magnetic structures in a broader range of materials.

ACKNOWLEDGMENTS

Appendix A: Polarization correction with *ex-situ* ^{3}He neutron spin filter

In this section, the polarization correction for measurements using an *ex-situ* ^{3}He-NSF as an analyzer without a spin flipper is described based on the polarization matrix method proposed by A. Wildes [26, 27]. The configuration of apparatus in this experiment is shown in Fig. 8. The spin state of incident neutrons polarized by the Cu_2MnAl single-crystal monochromator is defined as (+). The spin state of the scattered neutrons analyzed by a ^{3}He-NSFis defined as(+) or (−), where polarization can be flipped using AFP NMR. When the ^{3}He-NSF is polarized in the (+), the neutron intensity measured at the detector consists of four contributions as illustrated in Fig. 8. It is expressed as

$$\begin{aligned} I^{++} &= N_0 T_p p_p \Sigma^{++} T_a p_a \\ &+ N_0 T_p p_p \Sigma^{+-} T_a (1-p_a) \\ &+ N_0 T_p (1-p_p) \Sigma^{-+} T_a p_a \\ &+ N_0 T_p (1-p_p) \Sigma^{--} T_a (1-p_a), \end{aligned} \tag{A1}$$

where N_0 is the number of incident neutrons, T_p and T_a are the total neutron transmissions of the polarizer and analyzer, p_p and p_a are the polarization efficiencies of the polarizer and analyzer, Σ^{++}, Σ^{+-}, Σ^{-+}, and Σ^{--} are the scattering cross sections for the corresponding spin states. The polarization efficiencies, p_p and p_a are defined as $0.5 \le p_p \le 1$ and $0.5 \le p_a \le 1$, respectively. Their relation to the neutron polarization is given by $P_p = 2p_p - 1$ and $P_a = 2p_a - 1$. Only two configurations, I^{++} and I^{+-} can be measured in this experimental setup. Assuming that the numbers of incident neutrons with spin (+) and (−) from the reactor neutron source are equal and that the scattering cross sections satisfy $\Sigma^{++} = \Sigma^{--}$ and $\Sigma^{+-} = \Sigma^{-+}$, Eq. A1 can be rewritten in the following matrix form:

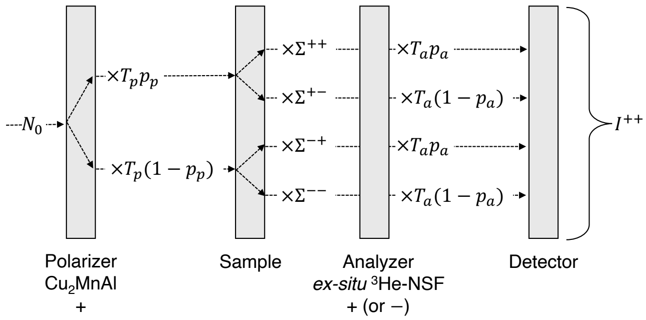

FIG. 8. Schematic diagram of the spin-dependent neutron intensity in the polarized neutron scattering experiment. The incident beam with intensity N_0 is polarized by the Cu_2MnAl polarizer and then scattered by the sample with the spin-dependent cross sections Σ^{++}, Σ^{+-}, Σ^{-+}, and Σ^{--}. The scattered neutrons are analyzed by the ^{3}He-NSF with transmission T_a and polarization efficiency p_a. The measured intensity I^{++} is given by the sum of the contributions from the four spin-scattering channels when Cu_2MnAl and ^{3}He-NSF are both polarized in the same direction.

$$\begin{bmatrix} I^{++} \\ I^{+-} \end{bmatrix} = \begin{bmatrix} T_p p_p T_a p_a + T_p(1-p_p)T_a(1-p_a) & T_p p_p T_a(1-p_a) + T_p(1-p_p)T_a p_a \\ T_p p_p T_a(1-p_a) + T_p(1-p_p)T_a p_a & T_p p_p T_a p_a + T_p(1-p_p)T_a(1-p_a) \end{bmatrix} \begin{bmatrix} \Sigma^{++} \\ \Sigma^{+-} \end{bmatrix}, \tag{A2}$$

Since the *ex-situ* ^{3}He-NSF exhibits a decay of the ^{3}He polarization over time, both the neutron polarization efficiency p_a and the neutron transmission T_a become time dependent. Accordingly, by treating Eq. A2 as a time-dependent matrix equation and taking its inverse, the following expression is obtained:

$$\begin{bmatrix} \Sigma^{++} \\ \Sigma^{+-} \end{bmatrix} = \frac{1}{T_p^2 T_a(t_1)T_a(t_2)P_p(P_a(t_1)+P_a(t_2))} \begin{bmatrix} T_pT_a(t_2)(1+P_pP_a(t_2)) & -T_pT_a(t_1)(1-P_pP_a(t_1)) \\ -T_pT_a(t_2)(1-P_pP_a(t_2)) & T_pT_a(t_1)(1+P_pP_a(t_1)) \end{bmatrix} \begin{bmatrix} I^{++}(t_1) \\ I^{+-}(t_2) \end{bmatrix}, \tag{A3}$$

where p_p and p_a are transformed into P_p and P_a using the relation described above. The experimentally determined values of $T_a(t)$, P_p, and $P_a(t)$ are described in the Sec. III A. While $T_a(t)$ and $P_a(t)$ vary with time owing to the relaxation of the ^{3}He polarization, T_p is constant throughout the measurement. Therefore, T_p only gives a common scale factor to the scattering cross sections. Thus, these polarization correction yields the non-spin-flip and spin-flip scattering cross sections, Σ^{++} and Σ^{+-}.

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